AI-Assisted Reversible Data Hiding Using Reinforcement Learning

Cheng Zhang, Bo Ou, Jun Yang, and Hua Deng

***Abstract*—Recently, artificial intelligence (AI) algorithm has been extensively utilized as an optimization tool in digital watermarking. However, existing works seldom consider the reversibility of the embedding framework, thereby neglecting the protection of sensitive carriers. In this paper, we propose an AI-assisted reversible data hiding (RDH) method based on reinforcement learning. In our method, the Q-Learning algorithm is introduced to address the optimization problem of RDH, i.e., adaptive two-dimensional (2D) mapping generation, and it is designed to simulate pixel modifications in 2D space, employing a Markov decision process formulation within the reinforcement learning paradigm. The new reward function is given to evaluate the effectiveness of 2D mapping based on the estimated embedding capacity and the distortion-capacity ratio. To ensure reversibility, a mapping adjustment strategy is implemented to update the environmental states. Experimental results show that the proposed method outperforms conventional 2D RDH and demonstrates competitive performance compared to other state-of-the-art RDH methods.**

***Index Terms*—Reversible data hiding, artificial intelligence, reinforcement learning, adaptive two-dimensional mapping.**

I. Introduction

ARTIFICIAL intelligence generated content (AIGC) technology has made significant progress in multiple fields, including text, images, and video, bringing unprecedented changes to multimedia content creation. However, with the popularization and application of AIGC, a series of new problems and risks have emerged, such as DeepFake, misinformation, and copyright infringement. Recently, watermarking-based traceability technologies have gradually become a focal point of interest for researchers and industry professionals, as they provide effective solutions to these security issues. Watermarking technology allows for the concealment of secret information within a digital carrier without compromising the quality of the carrier. The concealed information may include copyright details [1], [2], authentication fingerprints [3], [4], auxiliary data [5], and sensitive information [6].

Reversible data hiding (RDH) [7], [8], [9], [10] is one of the effective branches of watermarking. Compared to steganography [11] and robust watermarking [12], [13], [14], the advantage of RDH is that the original content of the carrier can be losslessly restored after the extraction of the secret information. This feature makes RDH a promising solution in scenarios where data integrity and precision are paramount. The necessity of reversibility is particularly evident in the following critical contexts: *1) Judicial Evidence*: When AIGC is used to reconstruct crime scenes, reversible watermarks (e.g., timestamps) can be embedded for authentication. Courtroom evidence requires pixel-perfect restoration of the original content to uphold legal integrity, making reversibility essential. *2) Medical Diagnosis*: In artificial intelligence (AI) assisted diagnostics, watermarks may be used to tag AI-generated medical images. To prevent any distortion from compromising diagnostic accuracy, the images must maintain pixel-level integrity. RDH ensures the original data can be recovered without affecting diagnostic value. *3) Cultural Heritage*: For digital archives such as Dunhuang murals, reversible watermarks can embed copyright information. Academic research on color and material science requires access to unaltered original data. RDH allows for watermark removal to guarantee research authenticity.

It is clear that in the above and similar scenarios, watermarking schemes causing permanent carrier distortion are unacceptable. This underscores the urgent need for a RDH framework that supports critical AIGC applications. In conventional RDH [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], secret information is embedded by extracting and utilizing redundant features of the carrier. These methods are designed based on domain expertise and experience. The embedding steps exhibit high interpretability, making them relatively easy to understand and explain. However, since manually designed features and optimization algorithms are often tailored for specific application scenarios, achieving robust generalization performance across different environments can be challenging. In addition, various manual strategy-based methods in the RDH field aim to improve embedding capacity or security. For example, the high-payload RDH scheme uses a weighted matrix to optimize embedding efficiency [29]; the dual-image-based RDH scheme employs three-pixel value difference expansion [30] or Hamming codes [31], [32] to achieve reversible hiding;

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the directional pixel value ordering (PVO) method combined with image interpolation enhances prediction accuracy [33]. There are also researches on using weighted matrices for reversible watermarking of color images [34], [35]. Regarding security, discrete cosine transform (DCT) steganography schemes based on weighted matrices [36], watermarking schemes based on local binary patterns [37], and Lagrange interpolating polynomial-based RDH [38] are employed for image authentication and tampering detection. These methods demonstrate the diversity of RDH but mostly rely on fixed rules or mathematical models, lacking adaptive optimization.

Recently, with the rapid advancement of AI, the use of machine learning and neural networks for embedding watermarking has yielded remarkable results, establishing itself as a mainstream direction in digital watermarking research. HiDDeN, proposed by Zhu et al. [39], is one of the pioneering schemes for embedding watermarks using deep neural networks (DNNs). In this method, the encoder and decoder networks are trained jointly to embed an invisible watermarking into the cover image. Ahmadi et al. [40] developed an end-to-end diffusion watermarking framework based on two fully convolutional neural networks (CNNs) with a residual structure. A differentiable network layer is utilized to simulate various attacks, thereby enhancing the security and robustness of the algorithm. Liu et al. [41] trained a style-transfer-based film-coating simulation network to improve robustness against film-coating effects on printed photographs. Fang et al. [42] proposed a decoupling noise layer for cross-channel simulation, which can achieve a general and robust watermarking network. Nadimpalli and Rattani [43] implemented a proactive DeepFake detection technique by employing generative adversarial networks (GANs) to embed visible watermarks at designated locations within generated fake images. Zeng et al. [11] designed a novel GAN-based watermarking method that incorporates an invisible backdoor. In their approach, imperceptible trigger samples are generated through steganography, and multiple backdoors are embedded into a GAN model to protect its intellectual property rights. Tan et al. [44] developed a template-based watermarking scheme for embedding anti-counterfeiting labels in ceramic patterns. This watermarking technique effectively resists distortion caused by decoration firing during the ceramic production process.

However, the aforementioned AI based watermarking methods result in permanent distortion of images. Considering the need for reversibility, it is challenging to implement an end-to-end RDH framework based on DNN. In the field of RDH, employing AI algorithms to optimize individual embedding steps is a common practice, such as the CNN-based pixel predictor [45], [46], [47], [48], [49], [50], [51], [52], the adaptive local complexity calculation using DNN [53], the optimal mapping path searching based on double deep Q-network (DDQN) [54], etc. These methods have demonstrated that deep learning algorithms significantly enhance RDH. Most existing research has concentrated on CNN-based pixel prediction, with limited exploration into optimizing adaptive embedding strategies.

In this paper, we propose an adaptive 2D modification-based RDH method utilizing the Q-Learning algorithm. Q-Learning [55], [56] is a classic reinforcement learning (RL) algorithm that has been validated in various decision-making contexts, including pattern recognition, natural language processing, and anomaly detection. The main contributions of this paper are summarized as follows.

1) An AI-assisted adaptive 2D histogram modification method is proposed to achieve more efficient reversible embedding. The mapping search path is transformed into a Markov decision process and an interactive training process is developed to gradually learn the expected cumulative return of each mapping action.
2) The Q-Learning algorithm is introduced into RDH. In addressing the optimization problem of 2D mapping and the performance evaluation in RDH, the components of reinforcement learning, such as action space, state transition, and immediate reward, are redefined. The optimal mapping can be derived from the learned experiences.
3) Extensive experiments have been conducted to demonstrate the effectiveness of the proposed method. The experimental results indicate that the 2D mapping generated using this method outperforms conventional RDH methods. When embedding an equal number of information bits, our approach achieves a higher peak signal-to-noise ratio (PSNR).

The remainder of this paper is organized as follows: In Section II, we introduce the key concepts of the AI-based RDH method and the Q-Learning algorithm. Subsequently, the proposed RDH method is described in detail in Section III. The experimental results are presented in Section IV. Finally, Section V concludes the paper.

II. Related Works

A. AI-Based RDH

First, we will provide a brief overview of the embedding method used in RDH. Generally, the information embedding process in RDH consists of three key steps (as illustrated in Fig. 1), i.e., pixel prediction, mapping generation, and pixel modification.

Initially, the cover pixels are predicted based on their neighboring pixels, allowing for the reorganization of pixels according to the difference between the cover pixel and its prediction, referred to as prediction errors (PEs). Next, the mapping relationships among the prediction errors are established based on the estimated embedding performance. The mapping "$a \rightarrow b$" indicates that a pixel with a PE value of a may be modified as b during the embedding process. Finally, the cover pixels are modified according to the generated mapping and the secret information. For the cover pixels (p_x, p_y) and the 2D mapping $\boldsymbol{M}_{x,y}$, where x and y represent the PE values of pixels p_x and p_y, respectively, the marked pixels will be obtained as follows

$$(p_x^*, p_y^*) = \begin{cases} (\hat{p}_x, \hat{p}_y) + \boldsymbol{M}_{x,y}^{(m)}, & \text{if } F(\boldsymbol{M}_{e_x,e_y}) > 1 \\ (\hat{p}_x, \hat{p}_y) + \boldsymbol{M}_{x,y}^{(0)}, & \text{else} \end{cases} \tag{1}$$

where $(\hat{p}_x, \hat{p}_y)$ represents the predicted values for the pixels (p_x, p_y). The notation $\boldsymbol{M}_{x,y}^{(k)}$ denotes the k-th mapping out-

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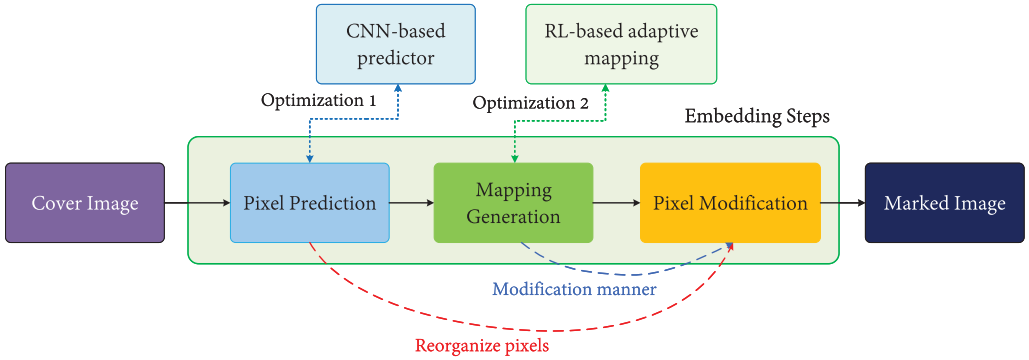


Fig. 1. The pipeline of information embedding in RDH.

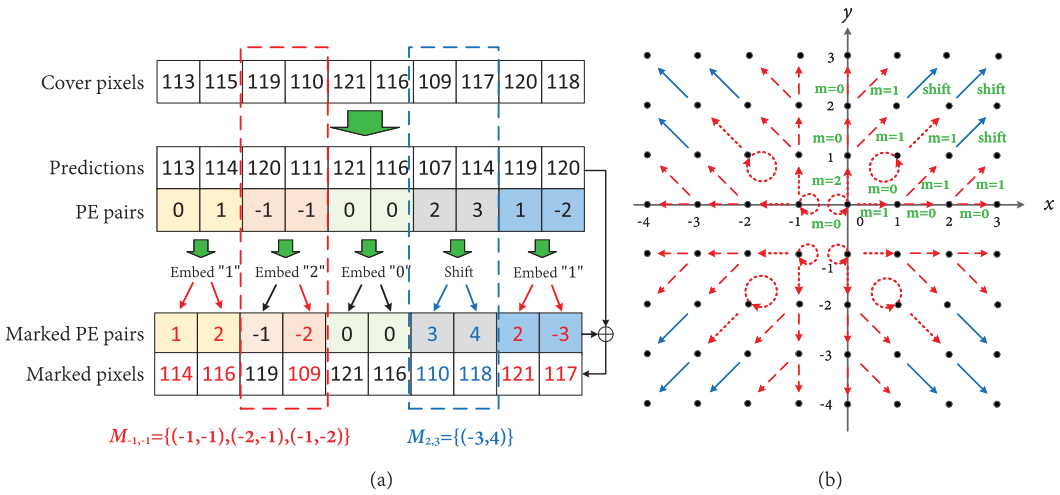


Fig. 2. The embedding and mapping of 2D RDH. (a) The steps of pixel modification according to the 2D mapping. (b) An example of the 2D mapping.

put for the PE pair (e_x, e_y). The variable m signifies the secret information, while $F(\boldsymbol{M}_{x,y})$ counts the total number of mapping outputs in the mapping $\boldsymbol{M}_{x,y}$. Specifically, the pixels (p_x, p_y) will be expanded to conceal secret information when their mapping yields multiple outputs. Conversely, they will either be retained or simply shifted. For example, in Fig. 2 (a), the mapping for the PE pair $(-1,-1)$ is given by $\boldsymbol{M}_{-1,-1} = \{(-1,-1),(-2,-1),(-1,-2)\}$. When the message to be embedded is $m = 2$, the marked pair is computed as $\boldsymbol{M}^{(2)}_{-1,-1} = (-1,-2)$.

For the time being, the most prevalent AI-based optimization in RDH involves utilizing CNNs to enhance pixel prediction. The greater the accuracy of the prediction, the more concentrated the pixel distribution will be after reorganization. This implies that a larger number of high-frequency elements can be employed to carry secret information, leading to increased embedding efficiency. To ensure reversibility, the primary challenge of prediction is to maintain the consistency of the predictor's input before and after the embedding process.

Different from conventional predictors [24], [57], [58], CNN-based methods can fully utilize global features to enhance local prediction results. In [46], Hu and Xiang introduced a novel pixel division method that classifies cover pixels into four subsets. During training, one subset is used as the label while the other three subsets serve as inputs to the network for generating predictions. This approach allows for the incorporation of more global information (three-quarters of the image) as context. They designed a feature extraction module utilizing several parallel convolution blocks and employed three convolution layers connected in series as the prediction module. Benefiting from multi-receptive fields and global optimization, the prediction performance is significantly improved. Subsequently, Yang and Huang added a convolution layer with a kernel size of 3×3 as a reconstruction module following the prediction module. Additionally, they proposed a square weighted mean filter consisting of a 5×5 neighborhood to construct the network input, further enhancing prediction accuracy. Rao et al. [51] developed a UNet-like network for the prediction task. Instead of using max pooling or average pooling, they employed a stride-2 convolution block that adaptively adjusts the weights of convolution kernels to select relevant information for downsampling feature maps. Furthermore, they suggested removing batch normalization layers to mitigate negative effects on the distribution of feature maps, thereby improving the network's predictive capability. Ma et al. [52] introduced a multi-scale fusion-enhanced interpolation network specifically designed for RDH, which integrates features across different scales to provide accurate reference pixels for subsequent predictions. Lin and Xiang

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[59] proposed enhancing the loss function by incorporating a smoothness factor. Training with this new loss function enables the predictor to make more accurate predictions in smooth areas.

In addition, optimizing mapping generation represents a promising area of research. As illustrated in Fig. 2 (b), the costs associated with embedding the same message can vary. For example, the distortion resulting from embedding the bit "1" into PE pair $(0, 0)$ is 1, whereas embedding it into the pair $(1, 1)$ results in a distortion of 2. The objective of mapping generation is to identify the optimal model that minimizes distortion while embedding a greater number of bits. Clearly, considering the frequency distribution complicates this optimization process. Unlike pixel prediction, mapping generation is a decision-making task, rendering DNN-based algorithms unsuitable for this application. For a given image, the optimal mapping result remains unknown, indicating a lack of supervised information during training. In light of this, we propose the development of an AI-assisted optimization method for 2D RDH based on the Q-learning algorithm.

B. Q-Learning

Reinforcement learning is a branch of machine learning that focuses on identifying optimal behavioral strategies through trial-and-error interactions with the environment. In continuous decision-making problems, such as navigating mazes, playing Atari games, or autonomous driving, the primary challenge is to determine the optimal action for each state. In reinforcement learning, a reward function and accumulated discounted return are utilized to train an agent to learn the appropriate action for a given state. Specifically, the optimization process aims to identify a strategy π that satisfies the following criteria,

$$\pi^* = \underset{\pi}{\operatorname{argmax}}\, \mathbb{E}(r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots) \tag{2}$$

where r_t denotes the immediate reward for the state s_t, $\gamma \in [0, 1]$ is the discount factor, and $\mathbb{E}(\cdot)$ represents the expectation function. When the agent encounters state s_t, it selects action a_t according to the strategy π and transforms to the state s_{t+1}. Theoretically, the optimal strategy enables the agent to choose actions that maximize the expected cumulative return. The challenge lies in accurately estimating this expectation. As a classic reinforcement learning algorithm, Q-Learning provides an effective approach to address this issue.

The core concept of Q-Learning is to learn a state-action value function, commonly referred to as the Q-function, which estimates the expected utility of taking a specific action in a given state. Grounded in the Bellman equation, the value estimation for each state-action pair is progressively refined by integrating immediate rewards with the predicted values of future states. Over time, the Q-function converges to the optimal value, enabling the agent to select the best action based on these values, thus achieving the learning objective. The updating rule for the Q-function is defined as follows

$$\begin{aligned} Q(s_t, a_t) \leftarrow\ & Q(s_t, a_t) \\ & + \alpha(r_t + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)) \end{aligned} \tag{3}$$

where $Q(s_t, a_t)$ represents the Q-function for the state-action pair (s_t, a_t), and $\alpha \in [0, 1]$ denotes the learning rate. Based on the final Q-function, the optimal actions are determined as

$$a_t^* = \underset{a}{\operatorname{argmax}}\, Q(s_t, a) \tag{4}$$

The primary advantages of Q-Learning include its simplicity, ease of implementation, and effectiveness in addressing problems with limited states, making it a suitable tool for solving 2D mapping optimization.

III. The Proposed Method

We propose to enhance the RDH method by adaptively generating a 2D mapping based on the Q-Learning algorithm. As illustrated in Fig. 3, an encoder G is trained to produce the optimal 2D mapping during the embedding process. The encoder takes the statistical characteristics H of the cover image and the required payload PS as inputs, and outputs the vectorized mapping result $\boldsymbol{M}$, i.e., $\boldsymbol{M} = G(H, PS)$. First, we will provide a brief introduction to the preprocessing steps.

A. Embedding Preprocessing

For a given grayscale image, let $(p_1, p_2, \ldots, p_N)$ represent the collected cover pixels. We utilize the rhombus predictor to calculate the predicted value as $\hat{p}_i = \lfloor (v_1 + v_2 + v_3 + v_4)/4 \rfloor$, where v_1, v_2, v_3, and v_4 are the four adjacent pixels located above, below, to the left, and to the right of p_i. Subsequently, we derive the PE pair sequence $(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_{N/2})$. Here, $\mathbf{e}_i = (e_{2i}, e_{2i+1})$ represents the combined adjacent PEs, and $e_{2i} = p_{2i} - \hat{p}_{2i}$ denotes the calculated PE value of p_{2i}. Next, the 2D prediction-error histogram (PEH) H is generated as

$$H_{x,y} = \#\left\{1 \le i \le \tfrac{N}{2} : \mathbf{e}_i = (x, y)\right\} \tag{5}$$

where # denotes the cardinality of a set, and $H_{x,y}$ represents the frequency distribution of PE pairs (x, y), the embedding performance of the current mapping can be estimated based on the 2D statistical characteristics. This estimation provides optimization feedback for the encoder. To facilitate adaptive embedding, the optimization problem can be defined as follows:

$$\boldsymbol{M}^* = \underset{\boldsymbol{M}}{\operatorname{argmin}} \frac{ED_{\boldsymbol{M}}}{EC_{\boldsymbol{M}}} \quad s.t.\ EC_{\boldsymbol{M}} \ge PS \tag{6}$$

where $ED_{\boldsymbol{M}}$ and $EC_{\boldsymbol{M}}$ represent the estimated embedding capacity (EC) and the embedding distortion (ED) of mapping $\boldsymbol{M}$ respectively.

In our method, the encoder consists of three components: the environment, the observation, and the agent. During each training epoch, the agent attempts to adjust the mapping based on the current strategy and observation state, while the environment subsequently evaluates its actions based on embedding performance. Utilizing this feedback, the agent can update its experience pool and modify its actions when encountering the same observation state again. Fig. 4 illustrates an example of the training steps. In the following sections, we will provide a detailed description of the method for generating the 2D mapping adaptively.

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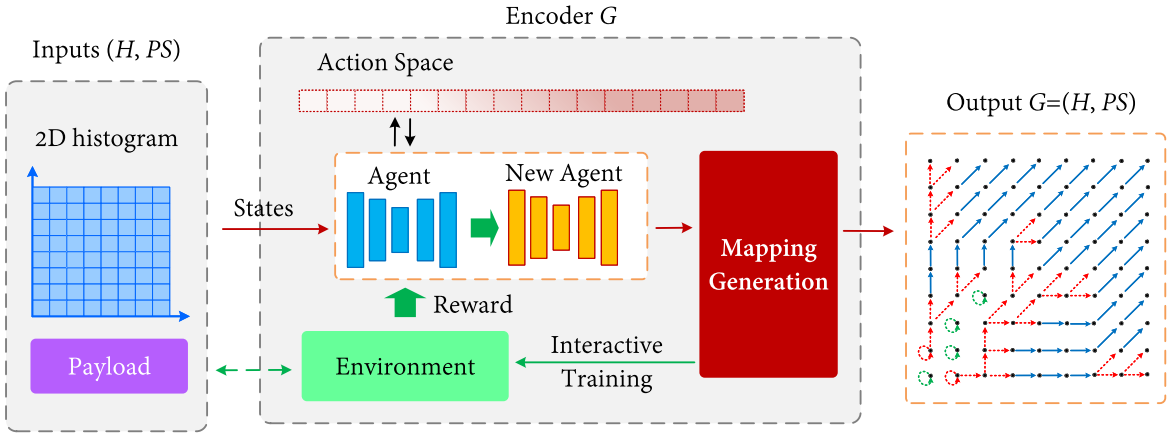


Fig. 3. The framework of the proposed method.

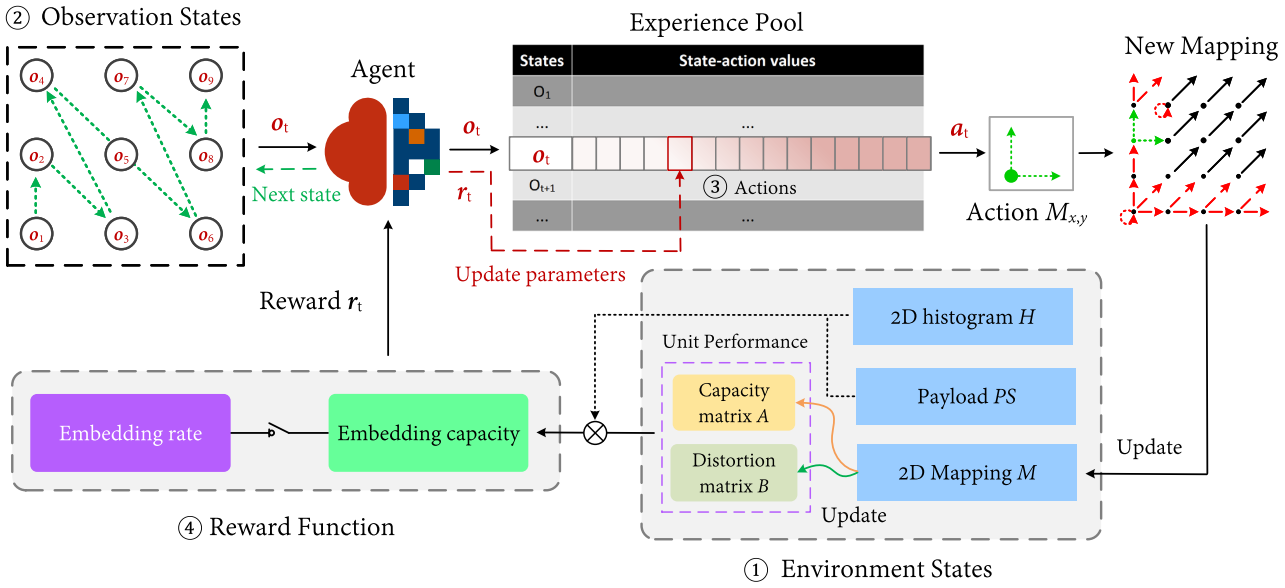


Fig. 4. An example of the interactive training system for optimizing the 2D mapping.

B. Training System

At the outset, we define a rectangular region of size $P \times P$ as the domain to be optimized. Since the mappings in the four quadrants are mirror images of one another, we will focus on the first quadrant as an example in the following sections. Specifically, only the mappings of the PE pairs that satisfy $x \le P-1$ and $y \le P-1$ will be adaptively determined, while the mappings for the other pairs will utilize a default model. Using a zigzag scanning order (i.e., (0,0) → (0,1) → (1,0) → (0,2) → (1,1) →...), the PE pairs to be optimized are examined sequentially to establish their mappings. This process optimizes the path $\{[(x_0, y_0), \boldsymbol{M}_{x_0,y_0}], [(x_1, y_1), \boldsymbol{M}_{x_1,y_1}],..., [(x_K, y_K), \boldsymbol{M}_{x_K,y_K}]\}$, where $K = P^2 - 1$. The encoder must continuously make decisions to achieve a globally optimal solution, which constitutes a Markov decision process. The mapping $\boldsymbol{M}_{x_t,y_t}$ depends solely on the current pair (x_t, y_t), without regard to previous decisions.

We employ Q-Learning to identify the optimal mapping path, which can be described as an interactive training system. This system consists of four key elements: the environment state, the observation state, the agent's action, and the reward function. These elements are defined as follows.

1) Environment State: The environment consists of the 2D histogram H, the specified payload PS, and the current 2D mapping state $\boldsymbol{M}$. Similar to conventional 2D RDH methods, the 2D mapping is initialized as

$$\boldsymbol{M}^{\dagger}_{x,y} = \begin{cases} \{(x,y),(x+1,y),(x,y+1) \\ ,(x+1,y+1)\}, & \text{if } x=0, y=0 \\ \{(x+1,y),(x+1,y+1)\}, & \text{if } x>0, y=0 \\ \{(x,y+1),(x+1,y+1)\}, & \text{if } x=0, y>0 \\ \{(x+1,y+1)\}, & \text{else} \end{cases} \tag{7}$$

According to the current mapping state $\boldsymbol{M}$, we can derive the initial unit capacity matrix A and the unit distortion matrix B by using Eqs. (8) and (9), respectively, i.e.,

$$A_{x,y} = \log_2 F(\boldsymbol{M}_{x,y}) \tag{8}$$

and

$$B_{x,y} = \frac{1}{F(\boldsymbol{M}_{x,y})} \sum_{(x',y') \in \boldsymbol{M}_{x,y}} |x' - x| + |y' - y| \tag{9}$$

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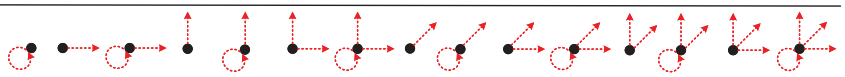
Mapping $M_{x,y}$															
Capacity $A_{x,y}$	0	0	1	0	1	1	$\log_2 3$	0	1	1	$\log_2 3$	1	$\log_2 3$	$\log_2 3$	2
Distortion $B_{x,y}$	0	1	1/2	1	1/2	1	2/3	2	1	3/2	1	3/2	1	4/3	1

Fig. 5. Fifteen possible mapping for the PE pair (x, y), and the corresponding results of unit capacity and unit distortion.

where $F(\boldsymbol{M}_{x,y})$ counts the total number of mapping outputs in the mapping $\boldsymbol{M}_{x,y}$. Meanwhile, the embedding performance result is calculated as

$$rate = \frac{ED_{\boldsymbol{M}}}{EC_{\boldsymbol{M}}} \tag{10}$$

Here, $EC_{\boldsymbol{M}}$ and $ED_{\boldsymbol{M}}$ represent the estimated embedding capacity (EC) and the embedding distortion (ED) of the mapping $\boldsymbol{M}$, respectively, i.e.,

$$\begin{cases} EC_{\boldsymbol{M}} = \sum(H \times A) \\ ED_{\boldsymbol{M}} = \sum(H \times B) \end{cases} \tag{11}$$

We define the environmental state as the set comprising a 2D mapping, performance metrics, and performance results, denoted as $s = \{\boldsymbol{M}, A, B, rate\}$. Upon receiving the action a_t from the agent, the original environmental state s_t is updated to produce the new state set s_{t+1}.

2) Observation State: The agent observes the environment and subsequently takes mapping actions. In our method, the observation is defined as the coordinates of a pair of points in the PEH, denoted as (x, y). For a domain of size $P \times P$, there are $K = P^2 - 1$ observation states, represented as $o_0, o_1, \ldots, o_K$. As mentioned earlier, the transition between states is fixed.

3) Agent's Action: For the observed state $o_t = (x, y)$, the agent selects an action $\boldsymbol{M}_{x,y}$ (i.e., a_t) from the action space. When the shifting step is set to 1, there are only four possible output directions for the PE pair (x, y), i.e., $(x, y) \to (x, y)$, $(x, y) \to (x + 1, y)$, $(x, y) \to (x, y + 1)$, and $(x, y) \to (x + 1, y + 1)$. Considering all combinations, the mapping $\boldsymbol{M}_{x,y}$ has $C_4^1 + C_4^2 + C_4^3 + C_4^4 = 15$ selections (see Fig. 5 for an illustration). Therefore, the action space can be defined as $\boldsymbol{\Theta} = (\theta_1, \theta_2, \ldots, \theta_{15})$.

4) Reward Function: The goal of Q-Learning is to identify an optimal path that maximizes the reward. Consequently, it is essential to design an appropriate reward function. Taking into account the optimization objective defined in Eq. (6), we define the reward function as follows,

$$r_t = \begin{cases} 3, & \text{if } EC_t \geq PS, rate_t < rate_{t-1} \\ 0, & \text{if } EC_t \geq PS, rate_t > rate_{t-1} \\ -1, & \text{if } EC_t < PS \end{cases} \tag{12}$$

where EC_t and $rate_t$ represent the optimal EC and embedding rate, respectively, calculated based on the state s_t. The design of Eq. (12) is driven by two core objectives for mapping optimization:

1) **EC-centric priority:** The primary constraint $EC_M \geq PS$ must be satisfied before minimizing the embedding rate. Therefore, assigning a reward of $r_t = -1$ when the payload demand is unmet strongly penalizes invalid actions, compelling the agent to first explore mappings with sufficient capacity. Conversely, the action receives positive feedback ($r = +1$) when the current 2D mapping meets the payload requirements.
2) **Directional guidance for optimization:** Once the payload requirement is met, a positive reward ($r_t = +2$) is given for an improved $rate_t$, i.e., a lower distortion per bit, while deteriorated performance ($r_t = -1$) penalizes inefficient explorations. This asymmetric reward structure accelerates convergence by consistently guiding the agent toward Pareto-optimal solutions in the distortion-capacity trade-off space.

Based on the definitions provided above, the optimization of 2D mapping involves identifying the optimal path $s_0, o_0, a_0, r_0, s_1, o_1, a_1, r_1, \ldots, s_K, o_K, a_K, r_K$. Utilizing a temporal-difference (TD) based algorithm, the encoder G can learn from experiences derived from incomplete episodes. Through iterative training, the mapping path can gradually converge.

C. Mapping Generation

According to Section III-B, for the state s_t and action a_t, the 2D mapping is updated as $\boldsymbol{M}^*_{x,y} = a_t$. It is evident that the agent's action does not consider the constraints of reversibility, i.e., 1) each PE pair has only one input, and 2) each PE pair has at least one output.

Only when the generated mapping is reversible is the optimization meaningful. The intuitive solution is to incorporate a penalty term into the reward function. However, our attempts have demonstrated that this approach complicates the training of the algorithm. In light of this, we adopt an alternative method, specifically mapping adjustment. For the states s_t, $o_t = (x, y)$, the mapping action a_t, and the previous mapping $\boldsymbol{M}_{x,y}$, suppose that (x_1, y_1) and (x_2, y_2) represent the new output and the old output, respectively, (x_3, y_3) denotes the input of (x_1, y_1), i.e., $(x_1, y_1) \in a_t$, $(x_2, y_2) \in \boldsymbol{M}_{x,y}$, and $(x_1, y_1) \in \boldsymbol{M}_{x_3,y_3}$, the rules for updating the 2D mapping $\boldsymbol{M}$ are outlined as follows (see Fig. 6 for an illustration).

- Case 1: If $F(\boldsymbol{M}_{x_3,y_3}) = 1$ and $F(a_t) = 1$, the mapping action is disabled. We keep the 2D mapping unchanged, i.e., $\boldsymbol{M}^*_{x,y} = \boldsymbol{M}_{x,y}$.
- Case 2: If $F(\boldsymbol{M}_{x_3,y_3}) = 1$ and $F(a_t) \neq 1$, the mapping action needs to be adjusted to avoid the conflict. We delete the output (x_1, y_1) from action a_t, i.e., $\boldsymbol{M}^*_{x,y} = a_t - (x_1, y_1)$.
- Case 3: If $F(\boldsymbol{M}_{x_3,y_3}) \neq 1$, the mapping state itself needs to be adjusted to avoid the conflict. That is, $\boldsymbol{M}^*_{x_3,y_3} = \boldsymbol{M}_{x_3,y_3} - (x_1, y_1)$, and $\boldsymbol{M}^*_{x,y} = a_t$.

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- Case 4: If $(x_2, y_2) \notin \boldsymbol{M}^*_{x,y}$, the mapping of PE pair (x_2, y_2) needs to be added the self-mapping to avoid the non-input situation, i,e., $\boldsymbol{M}^*_{x_2,y_2} = \boldsymbol{M}_{x_2,y_2} + (x_2, y_2)$, and $\boldsymbol{M}^*_{x,y} = a_t$.

Based on the aforementioned rules, the generated 2D mapping is always reversible. The detailed method for determining the optimal mapping path using Q-Learning is illustrated in Algorithm 1. This process consists of five phases. First, the training parameters are initialized. Second, using the ε-greedy policy, the mapping action a_t is selected for the observation o_t. Third, based on a_t and the previously mentioned adjustment rules, the 2D mapping is updated, and its performance is evaluated. Fourth, the reward is calculated based on the current environmental states, and the Q-function is updated accordingly. Finally, the optimal 2D mapping is derived from the Q-function after completing multi-epoch training.

Algorithm 1 Searching for the Optimal 2D Mapping With Q-Learning

Input : The $P \times P$ sized 2D histogram H, payload requirement PS, action space $\boldsymbol{\Theta}$, learning rate α, exploration rate ε, discount factor γ, maximal training epoch ξ, states $(o_0, o_1, ..., o_K)$, $K = P^2 - 1$.
Output: The optimal 2D mapping $\boldsymbol{M}^*$.
// Initialization
Initialize the 2D mapping as $\boldsymbol{M}^\dagger$, calculate the unit EC matrix $A^\dagger$ and unit ED matrix $B^\dagger$, initialize the performance result as $rate^\dagger$, initialize Q (Q-function) as zero.
// Train the agent
for $epoch \leftarrow 1$ **to** ξ **do**
  Set the termination marker $done = False$;
  Reset states: $\boldsymbol{M} \leftarrow \boldsymbol{M}^\dagger$, $A \leftarrow A^\dagger$, $B \leftarrow B^\dagger$, $rate \leftarrow rate^\dagger$, $o_t \leftarrow o_0$;
  while *not done* **do**
    // Select action a_t based on ε-greedy policy
    if $random(1) < \varepsilon$ **then**
      Select an action a_t randomly from $\boldsymbol{\Theta}$;
    else
      $a_t = \text{argmax}_a\, Q(o_t, a)$;
    end
    // Take action
    $\boldsymbol{M}_{x,y} \leftarrow a_t$; Adjust $\boldsymbol{M}$ to ensure the reversibility;
    Update A, B and search the best T to get $rate$; Get the reward r_t;
    if $o_t == o_K$ **then**
      // Termination state
      $Q(o_t, a_t) \leftarrow r_t$; done = True;
    else
      $Q(o_t, a_t) \leftarrow Q(o_t, a_t) + \alpha(r_t + \gamma \max_{a_{t+1}} Q(o_{t+1}, a_{t+1}) - Q(o_t, a_t))$;
    end
    // Go to next state
    $o_t \leftarrow o_{t+1}$;
  end
end
// Generate the optimal 2D mapping
for $(x, y) \leftarrow o_1$ **to** o_K **do**
  $o_t = (x, y)$; $\boldsymbol{M}^*_{x,y} \leftarrow \text{argmax}_a\, Q(o_t, a)$;
  Adjust $\boldsymbol{M}^*$ to ensure the reversibility;
end

D. Implementation Details

The processes of the proposed RDH method are briefly described in this section. For a given grayscale image, we first divide the pixels into two sets, respectively, "shadow" and "blank" (see [60], [61] for details). During the embedding process, both sets will be utilized to carry half of the secret bits. Since their operations are similar, we will use only the "shadow" set for illustration.

1) Preprocessing: First, the pixels are preprocessed to prevent overflow and underflow. Specifically, we modify the pixel value of "0" to "1" and the pixel value of "255" to "254". Meanwhile, a location map (LM) is employed to indicate these modifications, facilitating their identification during extraction. The pixels are then collected in a zigzag order to obtain the cover sequence $(p_1, p_2, \ldots, p_N)$. Next, we utilize the rhombus predictor to calculate the predicted value $\hat{p}_i$ and subsequently derive the PE pair sequence $(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_{N/2})$.

2) Embedding Optimization: Based on Algorithm 1, we can search for the optimal 2D mapping, denoted as $\boldsymbol{M}$, for the generated PEH. It is important to note that the pixel sorting technique is employed to further enhance the embedding. For the PE pair $\mathbf{e}_i$, a complexity measurement n_i (the calculation method can be referenced in [61]) is computed to determine the embedding priority. During the optimization process, we will traverse multiple thresholds to generate the corresponding 2D PEH. Specifically, Eq. (5) is redefined as

$$H^T_{x,y} = \#\left\{1 \le i \le \tfrac{N}{2} : \mathbf{e}_i = (x, y), n_i \le T\right\} \tag{13}$$

where T denotes the complexity threshold, the optimization objective is then modified as follows,

$$(\boldsymbol{M}^*, T^*) = \underset{\boldsymbol{M}, T^*}{\text{argmin}} \frac{ED_{\boldsymbol{M}}}{EC_{\boldsymbol{M}}} \quad s.t.\ EC_{\boldsymbol{M}} \ge PS \tag{14}$$

Since the performance of candidate 2D mapping is quantified into value matrices using Eqs. (8) and (9), the optimization can be solved using matrix operations. For the optimal threshold T^*, only the PE pairs that satisfy the condition $n_i \le T^*$ will be modified during the embedding process.

3) Data Embedding: Using Eq. (1), the secret bits are embedded into the first L PE pairs to derive $(\mathbf{e}^*_1, \ldots, \mathbf{e}^*_L, \mathbf{e}_{L+1}, \ldots, \mathbf{e}_{N/2})$. Correspondingly, the marked pixel sequence $(p^*_1, \ldots, p^*_{2L}, p_{2L+1}, \ldots, p_N)$ can be obtained. Next, the least significant bits (LSBs) of the first S_{aux} pixels are replaced with the bits of auxiliary information, where S_{aux} represents the size of the auxiliary information. Finally, the original LSBs are embedded back into the unprocessed pixels to derive the sequence $(p^*_1, \ldots, p^*_{2L}, p^*_{2L+1}, \ldots, p^*_{N_{end}}, p_{N_{end}+1}, \ldots, p_N)$. Note that, to ensure reversibility, the auxiliary information must include the compressed LM and its length, as well as the index N_{end} of the last processed PE pairs, the threshold T, and the compressed data of the 2D mapping M^*.

4) Data Extraction: For the marked image, we first collect the pixels and derive the PE pairs. The prediction aligns with the embedding process. Next, auxiliary information is extracted by reading LSBs of the first S_{aux} pixels. Based on the reconstructed 2D mapping, the original LSBs and the embedded secret bits are extracted from the marked sequences $(p^*_{N_{end}}, \ldots, p^*_{2L+1})$ and $(p^*_{2L}, \ldots, p^*_1)$ respectively. Meanwhile, the cover pixels are also recovered. Finally, the LSBs of the

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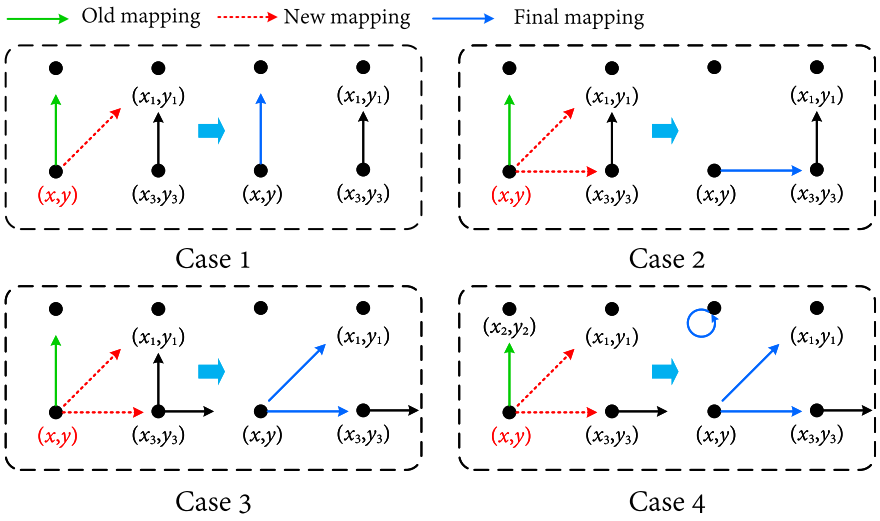


Fig. 6. Examples of the mapping adjustment rules.

TABLE I
PARAMETER SETTINGS OF THE PROPOSED METHOD

Parameter	Value
The size of the to-be-optimized region	8×8
Learning rate α	0.1
Discount factor γ	0.8
Exploration rate ε	0.1
Training epoch	3000

first S_{aux} pixels are restored, and any overflow or underflow pixels are corrected according to the LM.

IV. Experimental Results

In this section, we conduct several experiments to demonstrate the performance of the proposed method. The dataset employed includes three color images generated by AIGC model (FLUX.1), six standard images of size 512×512 from the USC-SIPI database, 24 test images from the Kodak database, and 1000 images from the BOSW-2 database. To verify the validity of the proposed method, we compare it against six advanced and representative baseline methods: pairwise prediction-error expansion (PEE) [60], skewed histogram shifting (HS) [23], multiple histogram modification (MHM) [24], adaptive pairwise PEE (APPEE) [61], CNN based predictor (CNNP) [62], and CNN-assisted RDH (CNNR) [46]. The parameter settings used in the proposed method are detailed in Table I (more explanations can be found in Section IV-D). The source code for our proposed scheme is available on GitHub at https://github.com/czhangkol/QLearningRDH.

A. Functional Demonstration

First, the embedding results are presented in Fig. 7. Utilizing the FLUX.1-Dev model (Raphael AI), we generated three images, each with dimensions of 512×512 pixels. Following this, we applied the proposed RDH method and MHM [24] respectively to embed additional 30,000 bits into each color channel of the test images. The proposed method modifies fewer pixels within the same region, indicating a more effective embedding strategy. Specifically, it often alters fewer pixels or leaves them unchanged altogether, demonstrating higher embedding efficiency. This approach effectively reduces image modifications, i.e., increases the payload per pixel, while embedding the same amount of secret data. The marked images generated by the proposed method maintain a high level of visual fidelity compared to the original images, achieving an average PSNR of 61.30 dB, which surpasses the 57.04 dB achieved by MHM. This elevated PSNR value signifies that the marked images are difficult to detect, thereby enhancing the imperceptibility of the embedded information and effectively concealing the embedding process. Furthermore, the reduced traces of modification contribute to improved security of the embedded content.

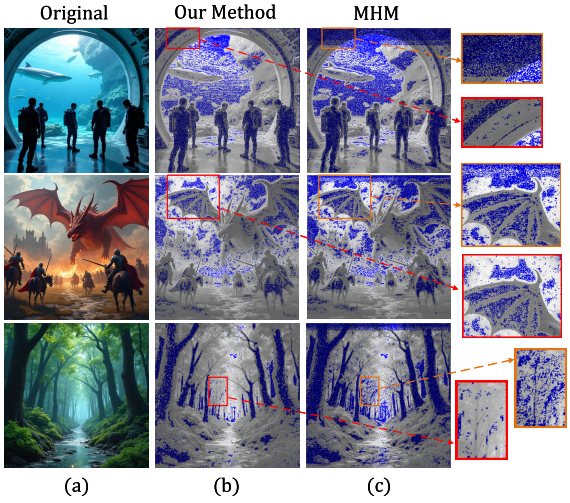


Fig. 7. The embedding results of the proposed method and MHM [24] are shown for three AIGC test images, where (a) represents the original images, and (b) and (c) depict the differences between the original and marked images.

B. Analysis of the 2D Mapping Generation

In this section, we analyze the proposed Q-Learning based optimization algorithm and the generated 2D mapping, using

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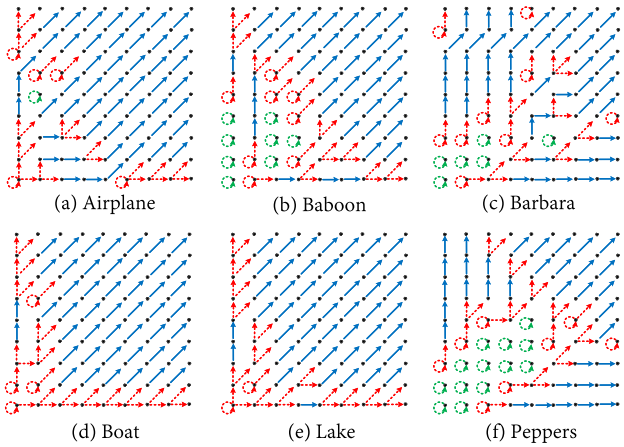


Fig. 8. The optimal 2D mappings generated using the proposed method for the first layer embedding in the images, where EC = 10,000 bits.

the USC-SIPI database. Initially, we analyze and test the generated 2D mappings. The key contribution of our method is that the 2D mapping can be adaptively determined based on the content of the image. Fig. 8 illustrates the adaptive results. We observe that the proposed method offers more effective variations for the embedding of pairwise PEE. The classification of unchanged bins, expansion bins, and shifting bins becomes more flexible, allowing for modifications that are better suited to different embedding requirements.

Next, we examine the training process of the proposed method. For a given capacity, Fig. 9 illustrates the average embedding rate curves and their standard deviation (SD) ranges for the 2D mappings generated during each epoch. It is evident that the embedding rate gradually decreases as the number of training epochs increases, ultimately converging steadily. At epoch 3000, the average embedding rates for the six test images are 0.57, 4.13, 1.20, 2.36, 1.74, and 2.37, respectively, while the SDs are ± 0.0003, ± 0.037, ± 0.008, ± 0.006, ± 0.008, and ± 0.04, demonstrating consistent convergence across independent trials. This indicates that the proposed method effectively optimizes the 2D mapping and achieves optimal results within a limited timeframe.

C. Performance Comparison

To better illustrate the embedding performance of the proposed method, we compare it with four representative adaptive RDH methods: the pairwise PEE method [60], skewed HS [23], MHM [24], and APPEE [61], using the Kodak and BOWS-2 databases, respectively. Additionally, we include CNNP [62] and CNNR [46] as baselines. These are recently proposed AI-assisted RDH methods. Notably, CNNP is an AI-based pixel prediction method. For convenience of comparison, we implemented its data embedding based on the traditional pairwise PEE method [60].

Firstly, we conduct a comparison using the Kodak database. For this experiment, the 24 test images are converted to grayscale and embedded with 10,000 bits and 20,000 bits, respectively. As shown in Fig. 10(a), the proposed method achieves the second-highest PSNR value for 17 images when the embedding capacity is set to 10,000 bits. Compared to pairwise PEE [60], skewed HS [23], MHM [24], APPEE [61], and CNNP [62], the average PSNR gains obtained by the proposed method are 0.81, 1.23, 1.18, 0.26, and 1.09 dB, respectively. Furthermore, when the embedding capacity is increased to 20,000 bits, the proposed method outperforms the five baseline methods, achieving average PSNR improvements of 0.87, 0.81, 0.74, 0.26, and 1.11 dB, respectively. This demonstrates the effectiveness of the proposed AI-assisted embedding strategy optimization algorithm. Here, skewed HS [23], MHM [24], and APPEE [61] employ different adaptive optimization strategies. The proposed method can export the pixel modification manner with higher efficiency. Compared to CNNP [62], the higher PSNR results of our method indicate that, under the same conditions, AI-assisted embedding strategy optimization achieves better results than AI-based pixel prediction optimization. However, the embedding performance of the proposed method is slightly lower than that of CNNR [46], which simultaneously performs AI-based pixel prediction and adaptive embedding optimization. Notably, the proposed method still demonstrates good comparability. The overall PSNR gap is mostly maintained below 0.5 dB, and our method performs better on specific images.

Then, we compare the average embedding performance of the proposed method with that of five baseline methods by varying EC from 5,000 bits to its maximum, using a step size of 5,000 bits. This comparison is conducted on the first 1,000 test images selected from the BOWS-2 database. The average PSNR values for the 1,000 images at different embedding capacities are plotted in Fig. 11. It is evident that the average PSNR results of the proposed method are consistently higher than those of the other five state-of-the-art methods. The PSNR gain achieved by the proposed method ranges from 0.1 to 1.2 dB, demonstrating its effectiveness in processing various types of images.

D. Ablation Studies

1) Ablation Study on 2D Mapping Optimization: As described above, an encoder is trained to optimize the mapping. To demonstrate its importance, we conducted control experiments without the optimization module and present the corresponding results in Table II (labeled with †). Here, we embed 10,000 bits and 20,000 bits, respectively, into each test image from the Kodak and BOWS-2 datasets, and calculate the PSNR between the cover image and the marked image. Clearly, the mapping optimized by the proposed encoder significantly outperforms the fixed mapping. The average PSNR value increases by 0.41 dB to 0.87 dB, demonstrating the effectiveness of the Q-Learning based mapping optimization algorithm.

2) Ablation Study on Key Hyperparameters: As shown in Table I, the proposed method relies on several parameters. Firstly, we treat the learning rate α as a variable and compare the embedding performance. Here, 20,000 bits of additional information are embedded into each test image from the Kodak dataset. The results are shown in Fig. 12(a). It can be observed that the PSNR results of the proposed method are similar

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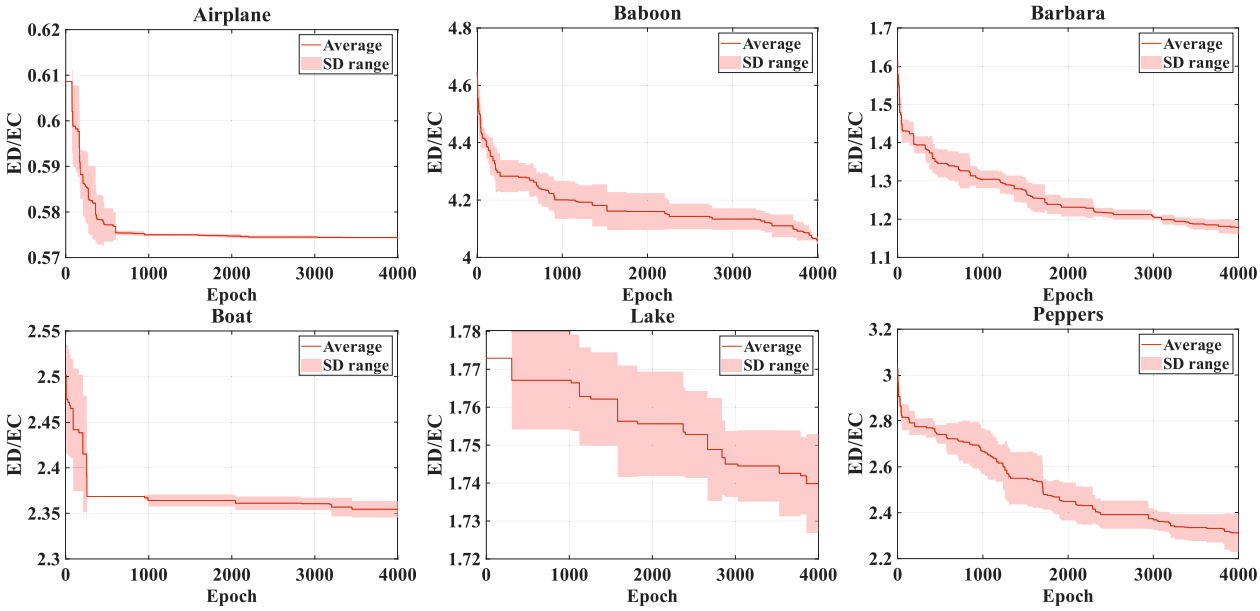


Fig. 9. The average embedding rate curves over five independent trials for six test images from the USC-SIPI database, embedding 5,000 bits in the first layer. All generated 2D mappings meet the capacity requirements.

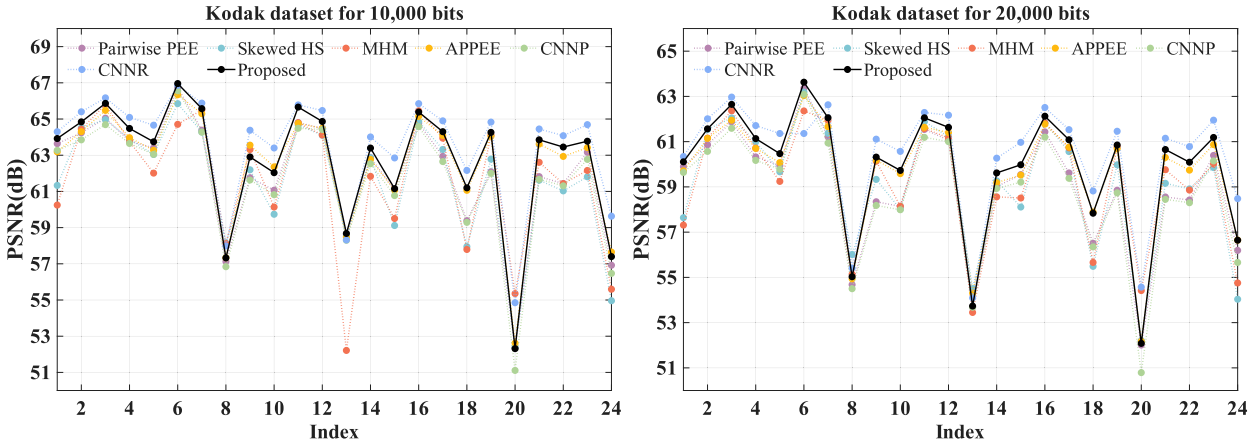


Fig. 10. A comparison of the PSNR (dB) between the proposed method and the pairwise PEE method [60], skewed HS method [23], MHM method [24], APPEE method [61], CNNP method [62], and CNNR method [46] is presented for the Kodak images, with embedding capacities of 10,000 bits and 20,000 bits.

TABLE II

THE AVERAGE PSNR RESULTS (IN DB) OF THE MARKED IMAGES AFTER EMBEDDING 10,000 BITS AND 20,000 BITS RESPECTIVELY TO THE 24 KODAK TEST IMAGES AND THE 10,000 BOWS-2 TEST IMAGES. † MEANS THE RESULTS WITHOUT MAPPING OPTIMIZATION

Dataset	EC (bits)	Skewed HS [23]	MHM [24]	APPEE [61]	CNNP [62]	CNNR [46]	Proposed†	**Proposed**
Kodak	10,000	61.58	61.62	62.55	61.70	63.58	61.99	**62.81**
	20,000	59.03	59.10	59.57	58.72	60.43	58.97	**59.84**
BOWS-2	10,000	62.98	62.13	62.96	63.07	-	63.14	**63.69**
	20,000	60.49	60.41	60.51	60.96	-	60.80	**61.21**

across different learning rates. For the parameter α, smaller values lead to better performance stability but at the cost of slower convergence. To balance these factors, we have chosen a learning rate of 0.1.

Next, we evaluate the embedding performance of the proposed method by varying the discount factor γ. We select three commonly used values (0.7, 0.8, and 0.9) and two additional values (0.1 and 0.5). According to Fig. 12(b), the proposed method generally performs better with a higher discount factor. Considering stability, setting $\gamma = 0.8$ is reasonable.

E. Computational Complexity Analysis

The time complexity of the proposed Q-Learning-based RDH method comprises two main components: Q-table updates and embedding execution. In each training epoch,

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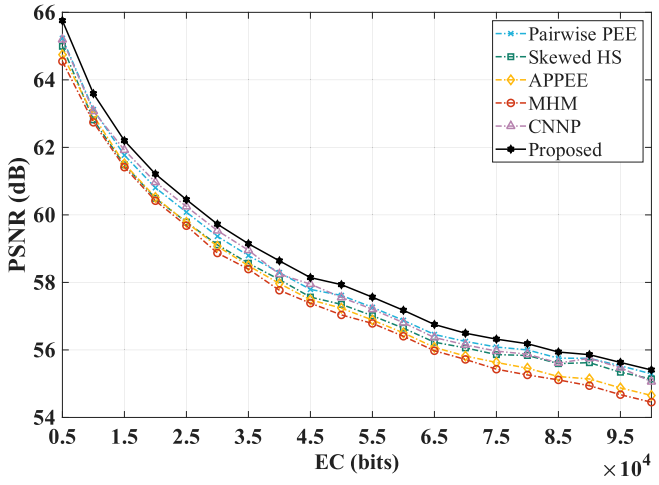


Fig. 11. The average PSNR values (in dB) for the first 1,000 test images from the BOWS-2 database at various embedding capacities are obtained using the proposed method, pairwise PEE [60], skewed HS [23], MHM [24], APPEE [61], and CNNP [62].

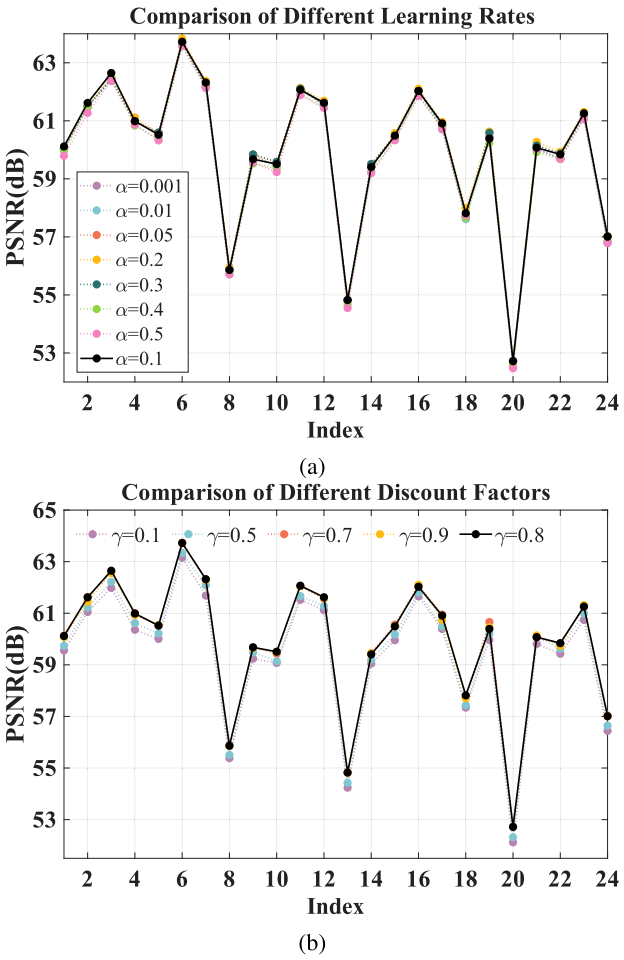


Fig. 12. Performance comparison of the proposed method with different hyperparameters on the Kodak database, using capacities of 20,000 bits.

we select one state-action pair per state to update its Q-value $Q(s, a)$. For an 8×8 optimization region, the total complexity is approximately $O(3 \times 10^6)$ after 3000 epochs. Compared to end-to-end DNN methods, the proposed approach eliminates the need for pre-training and avoids GPU-intensive backpropagation, although it requires retraining the agent for each image during application. Overall, the computational complexity of our method is acceptable. Compared to traditional optimization methods, the Q-Learning framework inherently avoids their limitations, i.e.,

- vs. Dynamic Programming (DP): DP requires exhaustive state traversal with a complexity of $O(8^{30})$, rendering it impractical for high-dimensional mapping optimization. In contrast, Q-Learning approximates the Bellman optimality equation through sampled experiments (Eq. (3)), effectively reducing the computational complexity.
- vs. Greedy Algorithms: While greedy methods sequentially minimize local distortion, they neglect the long-term cumulative effects of distortion. Q-Learning addresses this limitation by incorporating discounted future rewards (denoted by γ in Eq. (2)), thereby ensuring globally coordinated pixel modifications.

TABLE III

TIME COST COMPARISON (IN SECONDS) OF THE PROPOSED METHOD AND BASELINE APPROACHES (I.E., PAIRWISE PEE [60], SKEWED HS [23], MHM [24], APPEE [61], AND CNNP [62]) FOR EMBEDDING 10,000 BITS INTO SIX STANDARD 512 × 512 TEST IMAGES

Method	Airplane	Baboon	Barbara	Boat	Lake	Peppers
Proposed	44.27	39.87	43.18	38.04	36.12	35.78
[60]	0.09	0.11	0.10	0.11	0.10	0.10
[23]	0.47	-	0.38	15.74	0.37	9.48
[24]	22.06	25.40	24.79	30.02	24.32	28.37
[61]	3.77	3.82	3.76	3.76	3.75	3.78
[62]	6.93	5.91	6.71	5.68	7.88	3.68

To evaluate practical efficiency, we compare the running times of the proposed method with those of baseline approaches. The experiments were conducted on a computer equipped with a 3.5 GHz Intel i5-4600KF CPU, an NVIDIA 5060Ti GPU, and 32 GB of RAM. The proposed method, pairwise PEE, skewed HS, MHM, and APPEE were executed using MATLAB 2025a, while CNNP was run using PyCharm 2025. According to Table III, the average runtime of the proposed method is 39.54 seconds. Although this is significantly longer than the comparison methods, it remains within an acceptable range. It is worth noting that while the embedding time for CNNP is under 7 seconds, training its predictor still requires a substantial amount of time.

V. CONCLUSION

In this paper, we propose an AI-assisted adaptive 2D RDH method by formulating the mapping search as a Markov decision process and solving it using the Q-Learning algorithm. The strength of our approach lies in its simplicity and efficiency: the 2D mapping optimization is performed over discrete states consisting of 64×15 state-action pairs. Q-Learning's tabular method enables direct value estimation without requiring DNN approximation, thereby avoiding unnecessary over-parameterization. Additionally, we employ an action adjustment strategy rather than modifying rewards to maintain reversibility. Experimental results demonstrate that the proposed method outperforms several state-of-the-art RDH schemes in terms of embedding efficiency and image quality.

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However, the practical deployment of such a precise, reversible system requires a discussion of its limitations. The tabular nature of our Q-Learning method, while efficient for the defined state space, may encounter scalability challenges with extremely high-resolution images, as the state space could expand significantly. Furthermore, the method's primary strength—reversibility—also entails an inherent sensitivity to distortions. Like most RDH schemes, it is not inherently robust against common image processing operations such as JPEG compression or additive noise, which compromise the pixel-level accuracy necessary for perfect recovery.

These considerations directly inform our future work. We plan to optimize the representation of environmental states and reversible rewards and to train the system using more advanced reinforcement learning methods such as PPO, DDPG, and DDQN. These improvements aim not only to enhance scalability and efficiency but also to explore the potential for achieving more diverse embedding performance, including emerging anti-detection and anti-distortion capabilities tailored to specific application scenarios. In summary, while the current method excels in controlled, integrity-preserved environments (e.g., medical archiving, AIGC authentication), it provides a solid and efficient foundation upon which more robust and scalable versions can be developed.

APPENDIX
CONVERGENCE ANALYSIS OF Q-LEARNING IN RDH

Convergence Analysis

The convergence of Q-Learning relies on two key conditions:

1) **State-Action Space Finiteness**: The state space $\mathcal{S}$ (PE pairs within $P \times P$ region) and action space $\mathcal{A}$ (15 mapping actions) are finite, satisfying the Markov Decision Process (MDP) requirement.
2) **Reward Boundedness**: The reward function r_t (Eq. (12)) is bounded ($r_t \in \{-1, 0, 3\}$), ensuring the Q-value updates remain stable.

Under these conditions, the Q-function converges to the optimal Q^* with probability 1 when:

$$\sum_{t=0}^{\infty} \alpha_t = \infty \quad \text{and} \quad \sum_{t=0}^{\infty} \alpha_t^2 < \infty, \tag{15}$$

where α_t is the learning rate (set to 0.1 in Table I). Our experiments (Fig. 9) show the embedding rate stabilizes after 3000 epochs, confirming convergence.

Moreover, the update rule of Q-value (Eq. (3)) is derived from the Bellman optimality equation:

$$Q^*(s_t, a_t) = \mathbb{E}\left[r_t + \gamma \max_{a_{t+1}} Q^*(s_{t+1}, a_{t+1}) \right], \tag{16}$$

which holds since our MDP satisfies:

- **Markov Property**: State transitions depend only on current (x_t, y_t) and action a_t.
- **Discount Factor** $\gamma = 0.8$ ensures future rewards are prioritized, aligning with long-term distortion minimization.

Risk of Suboptimal Solutions

Q-Learning may converge to suboptimal mappings due to:

- **Local Optima**: The reward function (Eq. (12)) is non-convex. To mitigate this, we employ ϵ-greedy exploration ($\epsilon = 0.1$), allowing random actions to escape local optima.
- **State-Action Coupling**: Adjacent PE pairs share overlapping mapping outputs (Fig. 6). Our adjustment rules (Cases 1-4) decouple conflicts by dynamically updating $M_{x,y}$, ensuring global reversibility.

Empirical results (Table II) show the proposed method achieves 0.41-0.87 dB PSNR gains over fixed mappings, indicating effective avoidance of suboptimal solutions.

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